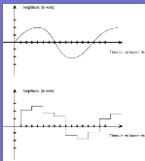


COM3502/4502/6502 SPEECH PROCESSING

Lecture 4

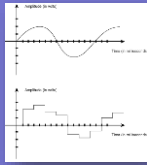
Digital Signals

Analogue vs. Digital Signals



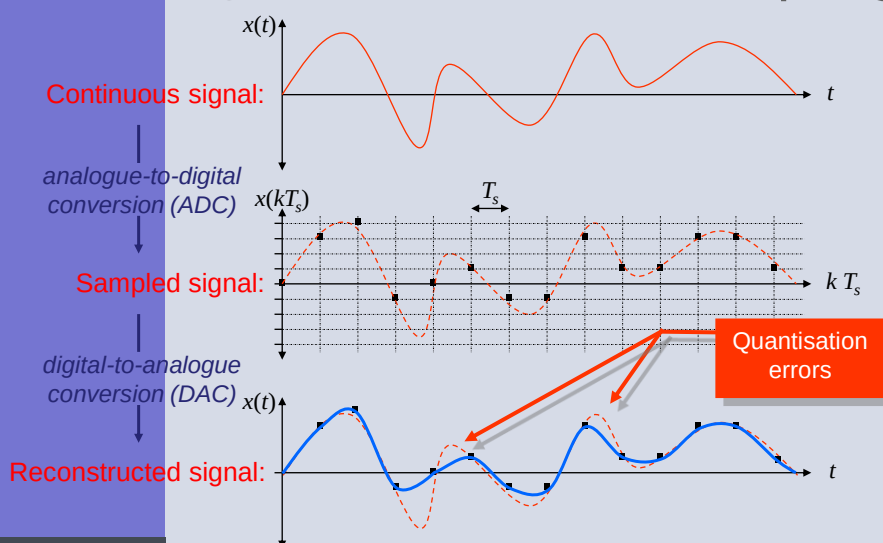
- All natural signals (*including speech*) are 'analogue' ...
 - they can assume an infinity of possible values
 - they exist continuously in time
- Perfect processing of analogue signals thus requires infinite resolution and/or infinite storage
- Any practical system (*computers or living organisms*) must sacrifice fidelity in order to process signals using finite resources
- Signals must therefore be 'quantised' in each dimension (amplitude and time)
 - Quantisation in time direction is called 'sampling'

Analogue vs. Digital Signals



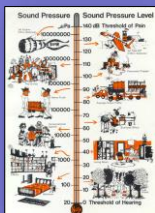
- Quantisation of a signal sequence in time direction is achieved by **'sampling'**
- Speech signals (e.g. from a microphone) are typically ...
 - quantised in amplitude
 - sampled in time
- The process of signal quantisation and sampling is known as: **'pulse code modulation' (PCM)**

Quantisation and Sampling

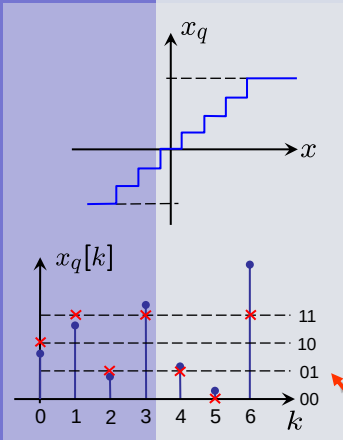


Amplitude Quantisation

- The range of numbers that can be used to represent a signal's amplitude defines the 'dynamic range' of the system
- If n bits are used to store each amplitude value, then 2^n possible values can be represented (0 to 2^n-1)
- The dynamic range is $20 \log_{10}(2^n-1)$ dB
- E.g. 16 bits of amplitude quantisation gives ...
 - 65535 possible values
 - 96 dB dynamic range



Quantization

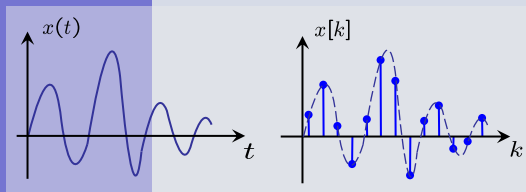


Bits	8	4	3
Music			
Speech			
Music with clipping			
Speech with clipping			

2 bit per value
(3 bit if negative values are considered)

Sampling of Continuous Signals

- Computers generally are not capable to deal with continuous signals
- Signals must be sampled and quantized



Sampling interval $\Delta T, T_s$

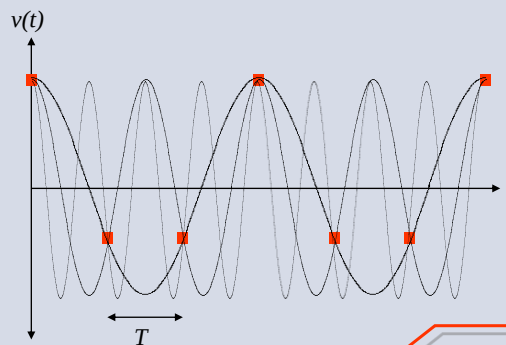
$$\Delta T = T_s = \frac{1}{f_s}$$

Sampling frequency f_s

$f_s = 8000\text{Hz}$ Speech

$f_s = 44100\text{Hz}$ Music /HiFi

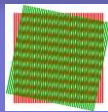
Sampling



Nyquist-Shannon theorem

To determine the frequency of a sampled sinusoid, you need *at least* two samples per period

Sample Rate

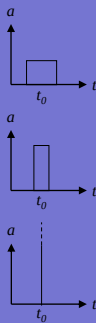


- The '**Nyquist-Shannon sampling theorem**' states that we need at least *twice* the number of sample points per second as the highest frequency in a signal ...
sampling rate $> 2 \times \text{max signal frequency}$
- Sampling a signal lower than the Nyquist rate leads to '**aliasing**' (energy at frequencies higher than the sampling rate are reflected back into the lower frequencies)
- E.g. aliasing can occur in digital images if the spatial frequencies are higher than the pixel resolution (the result is a '**Moiré pattern**')
- It is usual to low-pass filter a signal *before* sampling in order to avoid aliasing ...

$$f_c < 0.5 f_s$$

where f_c is the filter '**cut-off frequency**'
and f_s is the '**sampling frequency**'

Sampling Theory



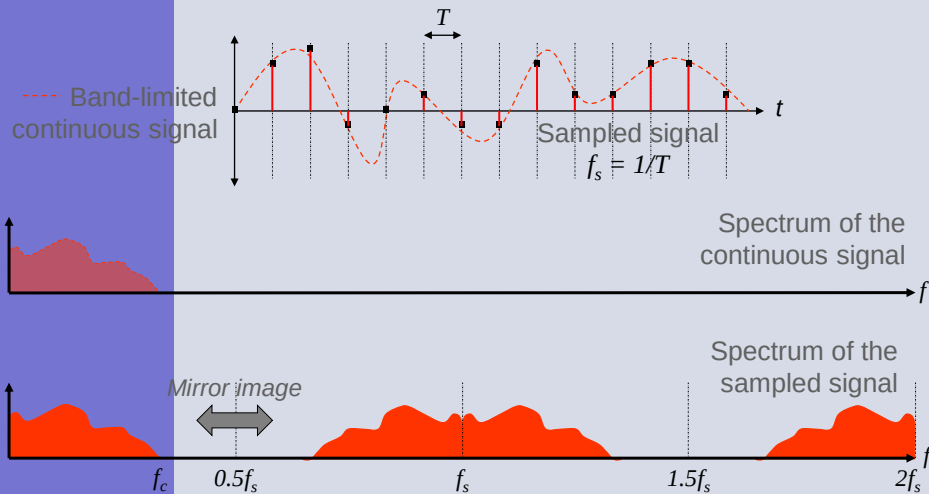
- Sampling a continuous signal $s(t)$ is equivalent to multiplying it by a '**Dirac delta function**'
- This impulse is a rectangle of the *unit area* centred around t_0 whose width tends to 0

$$\delta(t - t_0) = \begin{cases} \infty, & \text{if } t = t_0 \\ 0, & \text{if } t \neq t_0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

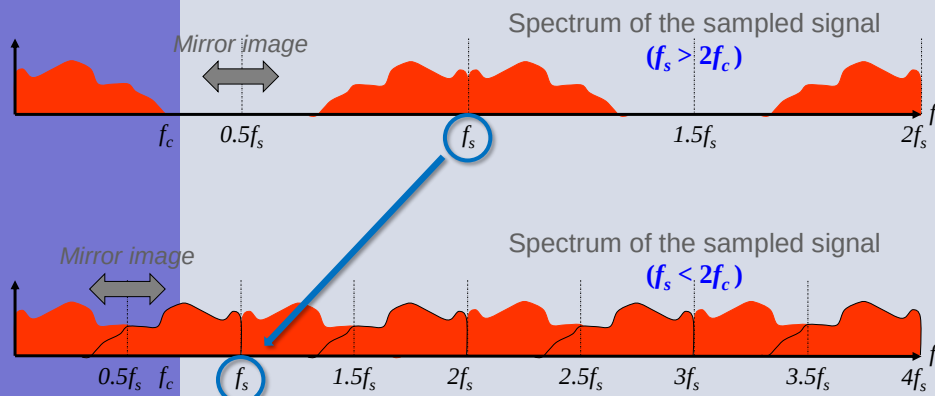
$$\int_{-\infty}^{\infty} s(t) \delta(t - t_0) dt = s(t_0)$$

Sampling Theory



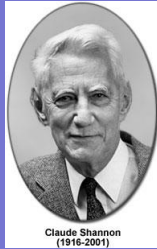
15

Aliasing



16

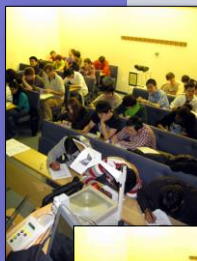
Coding Theory



Claude Shannon
(1916-2001)

- Derived from '**information theory**'
(founded in 1948 by Claude Shannon)
- Based on probability theory and statistics
- The most important quantities of information are ...
 - '**entropy**' (the information in a random variable)
 - '**mutual information**' (the amount of information in common between two random variables)
- Information is usually expressed in bits
 - entropy indicates how easily data can be '**compressed**' (due to redundancy)
 - mutual information can be used to find the communication rate through a channel

Digital Coding: *Images*



- Example image:
 - 360 pixels wide
 - 480 pixels high
 - 3 colours (*RGB*)
 - 1 byte/colour
- Raw data (*bitmap*):
 - 518.4 kilobytes ($360 \times 480 \times 3$)
- '**Lossless**' coding (*LZW-TIF*)
 - 378 kilobytes
- '**Lossy**' coding (*jpeg*)
 - 31 kilobytes

Digital Coding: *Images*



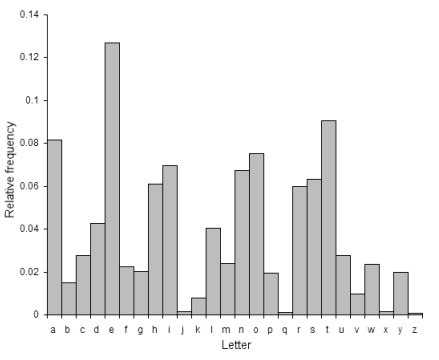
.tif



.jpg

Digital Coding: *Language*

International Morse Code



- 1. A dash is equal to three dots.
- 2. The space between parts of the same letter is equal to one dot.
- 3. The space between two letters is equal to three dots.
- 4. The space between two words is equal to seven dots.

A	••• —	U	•• —
B	••• —•	V	•• —•
C	•• —••	W	•• — —
D	•• —•	X	•• — —•
E	•••	Y	•• —••
F	•• —••	Z	•• — —••
G	•• —•		
H	•• —••		
I	•••		
J	•• —••		
K	•• —•	1	• — — — —
L	•• —•	2	•• — — —
M	•• — —	3	•• — —
N	•• —•	4	•• —•
O	•• — —	5	•• —
P	•• —•	6	•• —•
Q	•• —•	7	•• —•
R	•• —•	8	•• —•
S	•• —•	9	•• —•
T	•• —	0	•• —•

Digital Coding: *Signals*

- The amount of information needed to 'encode' a signal is defined by ...
 - amplitude quantisation (*in bits/sample*)
 - sampling rate (*in samples/second*)
- Digital signals are thus characterised in terms of their 'data rate' (*in bits/second - bps*)
 - ethernet LAN = 10 Gbps
 - wireless LAN = 600 Mbps
 - ADSL modem = 24 Mbps
 - 4G mobile data = ~10 Mbps
 - 3G mobile data = ~4 Mbps



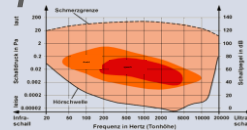
Note

b = bits

B = bytes

Digital Coding: *Speech*

- Speech has ...
 - a bandwidth of ~10 kHz
 - a dynamic range of ~50 dB
- Hence, the minimum quantisation and sampling requirements would seem to be ...
 - 20 kHz sampling rate
 - 8 bit quantisation
 - i.e. **160 kbps**
- However, it is possible to reduce both the bandwidth and dynamic range significantly before suffering a major drop in speech intelligibility
 - e.g. a telephone has a bandwidth of ~300 Hz to ~3.5 kHz (*which is a problem for sounds such as [f] and [s]*)
- Digital speech '**codecs**' make good use of lossy compression schemes (*by exploiting the 'source-filter' model of speech*)

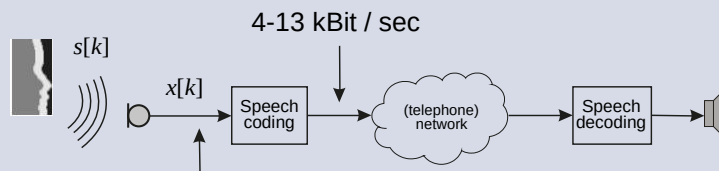


Note

b = bits

B = bytes

Digital Coding: *Speech*



- Digital telephone: $8000 \text{ Hz} * 8 \text{ Bit} = 64.000 \text{ Bit /sec}$ (ISDN)
- GSM (2G mobile): 4-13 kBit / sec

Audio/Speech Codecs

- DVD audio (24 bit 192 kHz PCM) = 4,608 kbps
- CD audio (16 bit 44.1 kHz PCM) = 705.6 kbps
- 16 bit 16 kHz PCM = 256 kbps
- telephone (8 bit 8 kHz ADPCM) = 64 kbps
- MP3 = 24 kbps
- mobile phone (GSM CELP) = 13 kbps
- VOIP (low rate) = 8 kbps
- NATO vocoder (channel) = 2400 bps
- NATO vocoder (LPC10e) = 2400 bps
- NATO vocoder (MELP) = 1200 bps

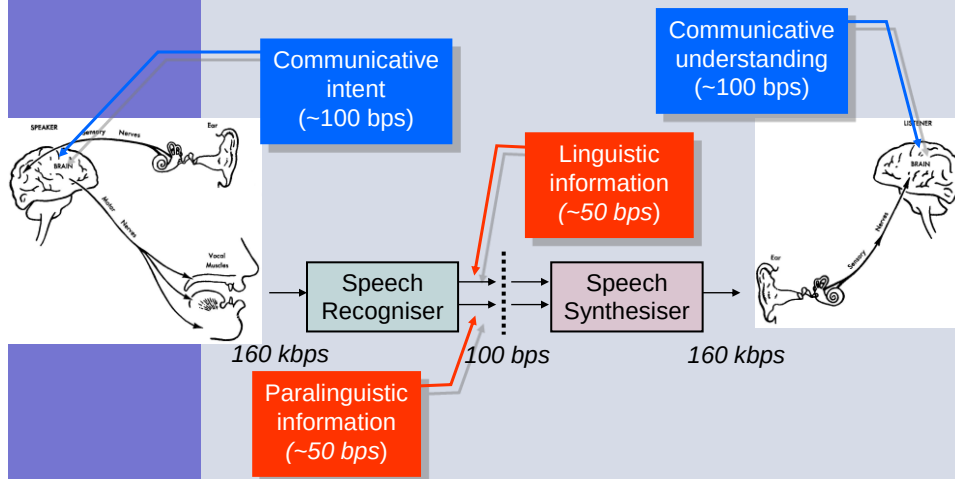
Note
b = bits
B = bytes

"vocoder" = "voice coder"

Digital Speech Coding

- The '**information rate**' in speech is estimated to be only ~100 bps!
 - linguistic information = ~50 bps
 - paralinguistic information = ~50 bps
- So why do we need kbps vocoders?
- The way to code signals at lower rates is to exploit any '**redundancies**' in the signal
- For speech, this is achieved using predictive models (*more on this in later lectures*)
- The ultimate predictive model for speech is '**speech recognition**' + '**speech synthesis**'

Very Low Rate Speech Coding



This slide set has covered ...

- Analogue vs. digital
- Quantisation and sampling
- A-to-D and D-to-A conversion
- Dynamic range
- Sampling theory
- Aliasing
- Coding theory
- Audio and speech coding

Any Questions ?

*Post in the Blackboard Discussion group
if you have questions later*



Next slide set ...

Waveform Processing